

- ◆ If you're using other functions such as interrupts and timers, turn them on.

Everything apart from this is similar to general C/C++ programming.

Writing an Arduino™ sketch...

Arduino™ programs called sketches are divided into two main parts (called functions):

- ◆ **setup** – In this part of the program, all the pin initialisation tasks are programmed, i.e. telling the microcontroller what task to perform.
- ◆ **loop** – After the appropriate task has been assigned to the corresponding pins and parts using setup, we need to run the code. All the code is put in the loop function. As evident in the name, the instructions in loop function are continuously repeated. This is done since the microcontroller is designed to repeat the same tasks again and again.

NOTE:

To use an external programmer, hold down [Shift] on your computer's keyboard when you hit the 'upload' button. The text will change to 'Upload using Programmer' and you can then upload using the external programmer.

Libraries

Libraries are files written in C or C++ (.c, .cpp) to provide extra functionality to your sketches. They lie at the crux of Arduino™ code and are the

STANDARD LIBRARIES	
Reads and writes to "permanent" storage.	EEPROM
Connects to the internet using the Arduino™ Ethernet shield.	Ethernet
Communicates with applications on the computer using a standard serial protocol.	Firmata
Connects to a GSM/GPRS network with the GSM shield.	GSM
Controls liquid crystal displays (LCDs).	LiquidCrystal
Reads and writes to SD cards.	SD
Controls servo motors.	Servo
Communicates with devices using the Serial Peripheral Interface (SPI) Bus.	SPI
Serial communication on any digital pins.	Software Serial
Controls stepper motors.	Stepper
Draws text images and shapes on the Arduino™ TFT screen.	TFT
Connects to the internet using the Arduino™ shield.	Wi-Fi
Two Wire Interface (TWI/I2C) that sends and receives data over a network of devices or sensors.	Wire.